Class 18: Pertussis Mini Project

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Background

Pertussis (aka whoopingh cough) is a common lung infection caused by the bacteria B. Pertussis.

This can infect all ages but is most severe for those under 1 year of age.

The CDC track the number of reported cases in US We can "scrape" this data with the **datapasta** package.

```
cdc <-
  data.frame(
                              year = c(1922,1923,1924,1925,1926,1927,
                                       1928,1929,1930,1931,1932,1933,1934,
                                       1935,1936,1937,1938,1939,1940,
                                       1941,1942,1943,1944,1945,1946,
                                       1947,1948,1949,1950,1951,1952,
                                       1953,1954,1955,1956,1957,1958,
                                       1959,1960,1961,1962,1963,1964,
                                       1965,1966,1967,1968,1969,1970,
                                       1971,1972,1973,1974,1975,1976,1977,
                                       1978,1979,1980,1981,1982,1983,
                                       1984,1985,1986,1987,1988,1989,
                                       1990,1991,1992,1993,1994,1995,
                                       1996,1997,1998,1999,2000,2001,
                                       2002,2003,2004,2005,2006,2007,
                                       2008,2009,2010,2011,2012,2013,
                                       2014,2015,2016,2017,2018,2019,2020,
                                       2021,2022,2023,2024,2025),
    cases = c(107473,164191,165418,152003,202210,181411,
                                       161799,197371,166914,172559,215343,
                                       179135,265269,180518,147237,214652,
                                       227319,103188,183866,222202,191383,191890,
```

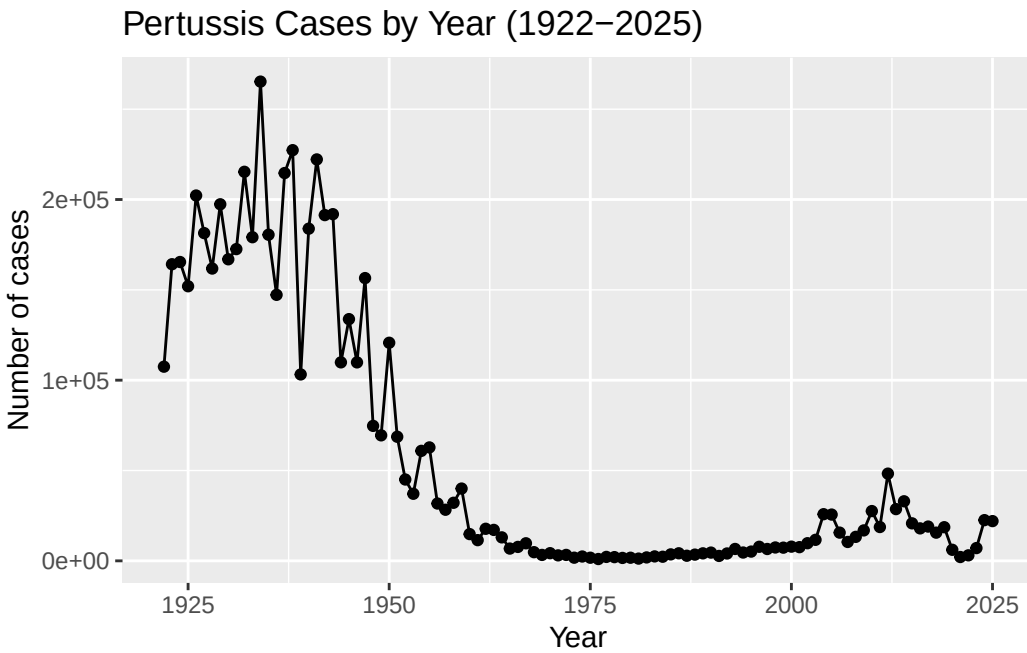
```
                                        109873,133792,109860,156517,74715,
                                        69479,120718,68687,45030,37129,60886,
                                        62786,31732,28295,32148,40005,14809,
                                        11468,17749,17135,13005,6799,7717,9718,
                                        4810,3285,4249,3036,3287,1759,2402,
                                        1738,1010,2177,2063,1623,1730,1248,
                                        1895,2463,2276,3589,4195,2823,3450,4157,
                                        4570,2719,4083,6586,4617,5137,7796,
                                        6564,7405,7298,7867,7580,9771,11647,
                                        25827,25616,15632,10454,13278,16858,
                                        27550,18719,48277,28639,32971,20762,
                                        17972,18975,15609,18617,6124,2116,3044,
                                        7063,22538, 21996)
  )
```

Q1. Make a plot of `year` vs. `cases`. With the help of the R "addin" package datapasta assign the CDC pertussis case number data to a data fram called cdc and use ggplot to make a plot of cases numbers over time.

```
library (ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
ggplot (cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs (title = "Pertussis Cases by Year (1922-2025)", x = "Year", y = "Number of cases")
```



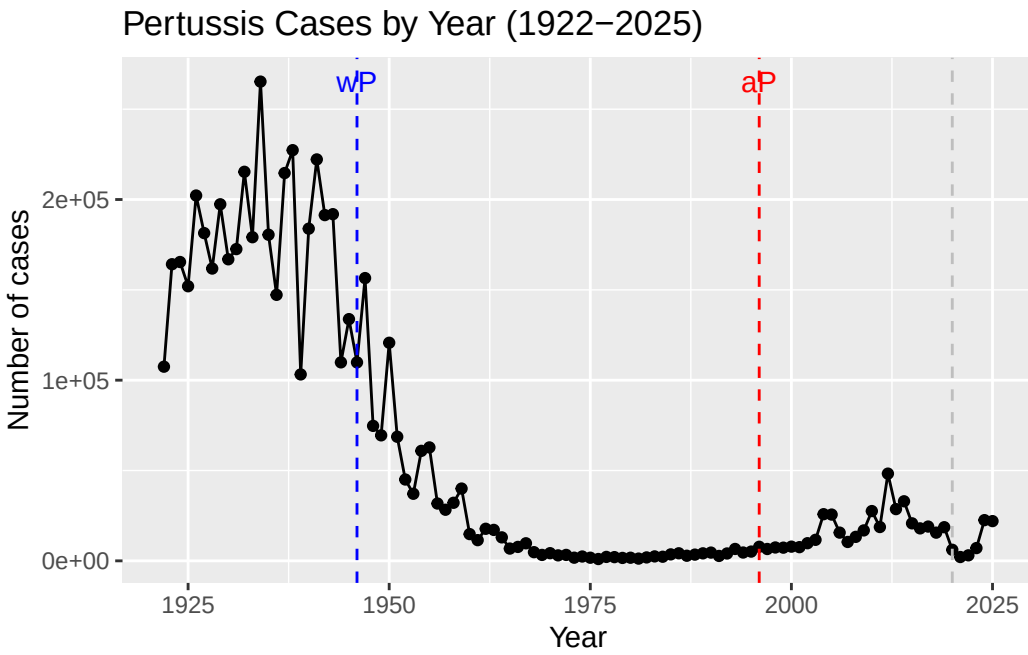
Q. Add some major milestones including the first wP vaccine roll-out (1946), the switch to the new aP vaccine (1996), the COVID years (2020)

Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

Before the introduction of the wP vaccine, pertussis cases were very high (usually greater than 100,000 per year). After the introduction and use of the wP vaccine, pertussis cases dropped dramatically to very low levels. But after the introduction and switch to the aP vaccine, pertussis cases started increasing slowly again but it is still lower than pre-vaccine levels.

```
library (ggplot2)

ggplot (cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs (title = "Pertussis Cases by Year (1922-2025)", x = "Year", y = "Number of cases") +
geom_vline(xintercept = 1946, col = "blue", lty=2) +
   geom_vline(xintercept = 1996, col = "red", lty=2) +
 geom_vline(xintercept = 2020, col = "gray", lty=2) +
annotate("text", x = 1946, y = max(cdc$cases), label = "wP", color = "blue") +
annotate("text", x = 1996, y = max(cdc$cases), label = "aP", color = "red")
```



for cases

22538 for year 2024 21996 for year 2025

> Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine, pertussis cases started increasing slightly. A possible explanation for this observed trend is that the aP vaccine provides weaker/shorter-lasting immunity compared to the wP vaccine. Due to the immunity fading over time, more people become as risk for infection which is why there is a rise in pertussis cases after 1996. Also people being more hesitant in regards to vaccinations could be a possible reason as to why there is a slow rise in pertussis cases.

There were high case numbers in the pre 1940s. Booster shot helps with decreasing the number of vaccine. wP better than aP vaccine.

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

#CMI PB project Computational Models of Immunity – Pertussis Boost project aims to provide the scientific community with this very information.

They make their data avilable via JSON format returning API. We can read this in R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)
```

```
Warning: package 'jsonlite' was built under R version 4.4.3
```

```
subject <- read_json("http://cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

There are 87 aP infancy vaccinated subjects in the dataset. There are 85 wP infancy vaccinated subjects in the dataset.

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

There are 60 male patients in the dataset. There are 112 female patients in the dataset.

```
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

> Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)?

In regards to American Indian/Alaska Native: 0 females and 1 male. In regards to Asians: 32 females and 12 males. In regards to Black or African American: 2 females and 3 males. In regards to more than one race: 15 females and 4 males. In regards to native Hawaiian or other pacific islander: 1 female and 1 male. In regards to unknown or not reported: 14 females and 7 males. In regards to white: 48 females and 32 males.

```
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

> Q. In terms of race and gender, is this dataset representative of the US population?

```
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

The average age of wP individuals is 28. The average age of aP individuals is 37. They are significantly different since the individuals who got the wP vaccine are generally older and the difference between the mean ages of the two groups is about 9 years which is a large difference.

```r
library (lubridate)
```

```
Warning: package 'lubridate' was built under R version 4.4.3


Attaching package: 'lubridate'


The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```r
subject$age <- today() - ymd(subject$year_of_birth)
subject$age
```

```
Time differences in days
  [1] 14676 21251 15772 13946 12850 13946 16502 15041 11024 16137 14676 16137
 [13] 10658 12119 13580 14311 16868 10658 11754 16502 15772 15041 12850 12485
 [25] 13946 15772 10658 16137 10658 13946 13580 10658 13215 15772 12850 10658
 [37] 10293 10658 15041 11754 15041 10658 10293 10293 10658 10293 11024 10293
 [49] 10658 10658 10658 10293 10293 10658 10658 10658 11024 10658 10658 10658
 [61] 14311 12119 11389 12119 13215 18329 19790 19790 13215 10293 10293 12850
 [73] 11389 11389 10293 10293 13946 12119 14311 12485 12119 10293  9928 10658
 [85]  9563 10293  9563  9563 10658  9928 10293  9563 11024  9928 10293  9563
 [97] 14676 12119  9928  9197  8467  8467 11754 13580 11754 11024 10293 11389
[109] 13580 10658 11024 11024 11024 13215  8832  9563 11754 10293 10293 11389
[121]  9563  9928 11024  9563 12119 12119 11024 11754 12850 11024 10293 11389
[133] 10658 13215 11389 11389 10293  9563 12119  9197 11024 12850  8467  9928
[145]  8832 12485  9563 13946 12850 12850 12485 11389 10293 10658 10658  9197
[157] 10658  9563 11754 11024 12119  9928 12119 12850 12119  9197 10658 12850
[169]  8467 12485  8467 14676
```

```r
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
ap <- subject %>% filter(infancy_vac == "aP")

round( summary( time_length( ap$age, "years" ) ) )
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      27      28      28      29      35 
```

```r
# wP
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      33      35      37      40      58 
```

Q8. Determine the age of all individuals at time of boost?

The average age of all individuals at time of boost is 26.

```r
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

```r
age_at_boost
```

```
  [1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481 35.84942 34.14921
  [9] 20.56400 34.56263 30.65845 34.56263 19.56194 23.61944 27.61944 29.56331
 [17] 36.69815 19.65777 22.73511 35.65777 33.65914 31.65777 25.73580 24.70089
 [25] 28.70089 33.73580 19.73443 34.73511 19.73443 28.73648 27.73443 19.81109
 [33] 26.77344 33.81246 25.77413 19.81109 18.85010 19.81109 31.81109 22.81177
 [41] 31.84942 19.84942 18.85010 18.85010 19.90691 18.85010 20.90897 19.04449
 [49] 20.04381 19.90691 19.90691 19.00616 19.00616 20.04381 20.04381 20.07940
 [57] 21.08145 20.07940 20.07940 20.07940 32.26557 25.90007 23.90144 25.90007
 [65] 28.91992 42.92129 47.07461 47.07461 29.07324 21.07324 21.07324 28.15058
 [73] 24.15058 24.15058 21.14990 21.14990 31.20876 26.20671 32.20808 27.20876
 [81] 26.20671 21.20739 20.26557 22.26420 19.32375 21.32238 19.32375 19.32375
 [89] 22.41752 20.41889 21.41821 19.47707 23.47707 20.47639 21.47570 19.47707
 [97] 35.90965 28.73648 22.68309 20.83231 18.83368 18.83368 27.68241 32.68172
[105] 27.68241 25.68378 23.68241 26.73785 32.73648 24.73648 25.79603 25.79603
[113] 25.79603 31.79466 19.83299 21.91102 27.90965 24.06297 23.90965 27.12115
[121] 22.12183 23.12115 26.17933 22.17933 29.17728 29.23477 26.23682 28.29295
[129] 31.29363 26.29432 24.35044 27.35113 25.40999 32.41068 27.56194 27.41136
[137] 24.50650 22.56263 29.56057 21.69473 26.69678 31.90691 19.90691 23.90691
[145] 20.90623 31.00616 23.00616 35.00616 32.00548 32.00548 31.04449 28.12047
[153] 25.11978 26.11910 26.19302 22.19302 26.19302 23.19507 29.19370 27.32923
[161] 30.32717 24.55852 30.55715 32.55852 30.55715 22.67488 26.67488 32.67625
[169] 20.67625 31.75086 20.86516 36.06297
```

```r
summary(age_at_boost)
```

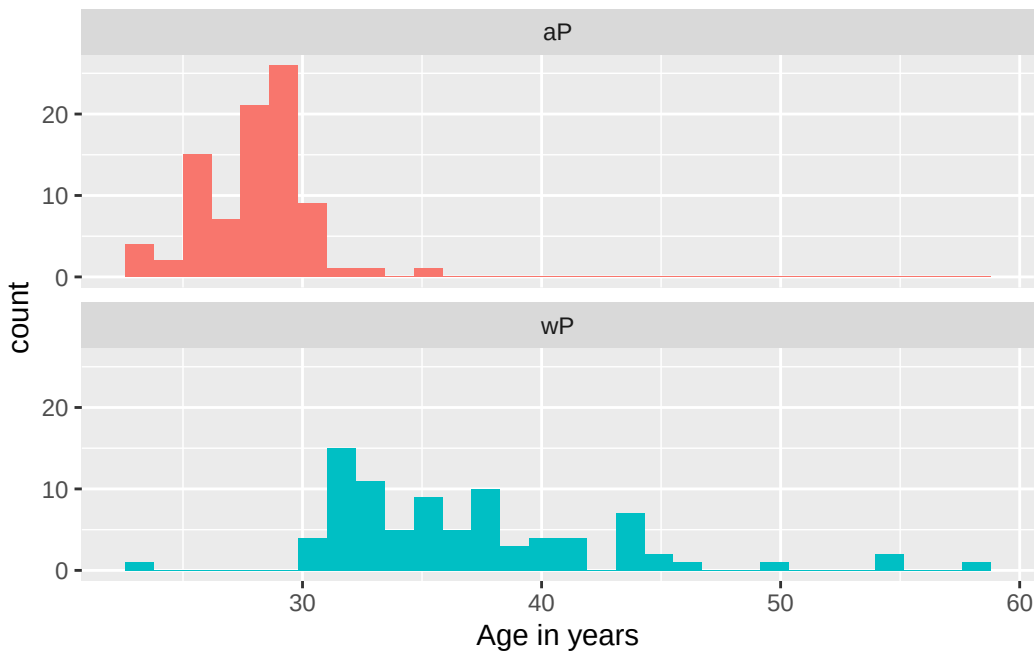
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  18.83   21.03   25.75   26.09   29.56   51.07 
```

> Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

Based on the histograms below, yes the two groups are significantly different. This conclusion is further supported due to the p-value test as the calculated p-value is 2.37e-23 which is less than 0.05 which means that the difference is statistically significant. Individuals who got the aP vaccine are generally younger while individuals who got the wP vaccine are generally older.

```r
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
# Or use wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
       time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

Joining multiple tables

Let's read some more database tables:

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)

ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

Completed

```
library (dplyr)

meta <- left_join(specimen, subject, by = "subject_id")
dim(meta)
```

```
[1] 1503   14
```

```
head(meta)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1          wP         Female
2                             1         Blood     2          wP         Female
3                             3         Blood     3          wP         Female
4                             7         Blood     4          wP         Female
5                            14         Blood     5          wP         Female
6                            30         Blood     6          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
         age
1 14676 days
2 14676 days
3 14676 days
4 14676 days
5 14676 days
6 14676 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

Completed

```
library (dplyr)

abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```
dim(abdata)
```

```
[1] 61956    21
```

```
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id actual_day_relative_to_boost
1 UG/ML                 2.096133          1                           -3
2 IU/ML                29.170000          1                           -3
3 IU/ML                 0.530000          1                           -3
4 IU/ML                 6.205949          1                           -3
5 IU/ML                 4.679535          1                           -3
6 IU/ML                 2.816431          1                           -3
  planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1          wP         Female
2                             0         Blood     1          wP         Female
3                             0         Blood     1          wP         Female
4                             0         Blood     1          wP         Female
5                             0         Blood     1          wP         Female
6                             0         Blood     1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
```

```
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
         age
1 14676 days
2 14676 days
3 14676 days
4 14676 days
5 14676 days
6 14676 days
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

There are 6698 specimens for the IgE isotype. There are 7265 specimens for the IgG isotype. There are 11993 specimens for the IgG1 isotype. There are 12000 specimens for the IgG2 isotype. There are 12000 specimens for the IgG3 isotype. There are 12000 specimens for the IgG4 isotype.

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

There are four different datasets which includes 2020, 2021, 2022, and 2023. The most recent dataset has the second most rows. This could be due to the scientists having more samples to process and there is more cases to analyze compared to the last two datasets.

```r
unique(abdata$dataset)
```

```
[1] "2020_dataset" "2021_dataset" "2022_dataset" "2023_dataset"
```

```r
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

Examine IgG Ab titer levels

Let's focus on the IgG antigen and make a plot of MFI_normalized for all antigens.

```r
igg <- abdata |>
  filter(isotype == "IgG")

head(igg)
```

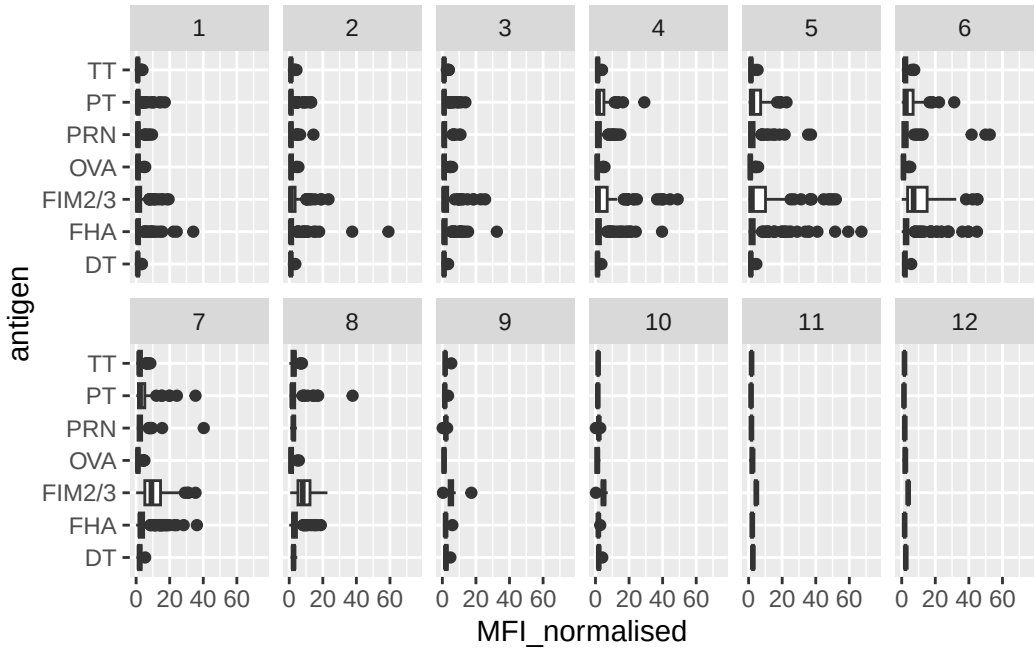
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id actual_day_relative_to_boost
1 IU/ML                 0.530000          1                           -3
2 IU/ML                 6.205949          1                           -3
3 IU/ML                 4.679535          1                           -3
4 IU/ML                 0.530000          3                           -3
5 IU/ML                 6.205949          3                           -3
6 IU/ML                 4.679535          3                           -3
  planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1          wP         Female
2                             0         Blood     1          wP         Female
3                             0         Blood     1          wP         Female
4                             0         Blood     1          wP         Female
5                             0         Blood     1          wP         Female
```

```
6                              0         Blood     1           wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
         age
1 14676 days
2 14676 days
3 14676 days
4 15772 days
5 15772 days
6 15772 days
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

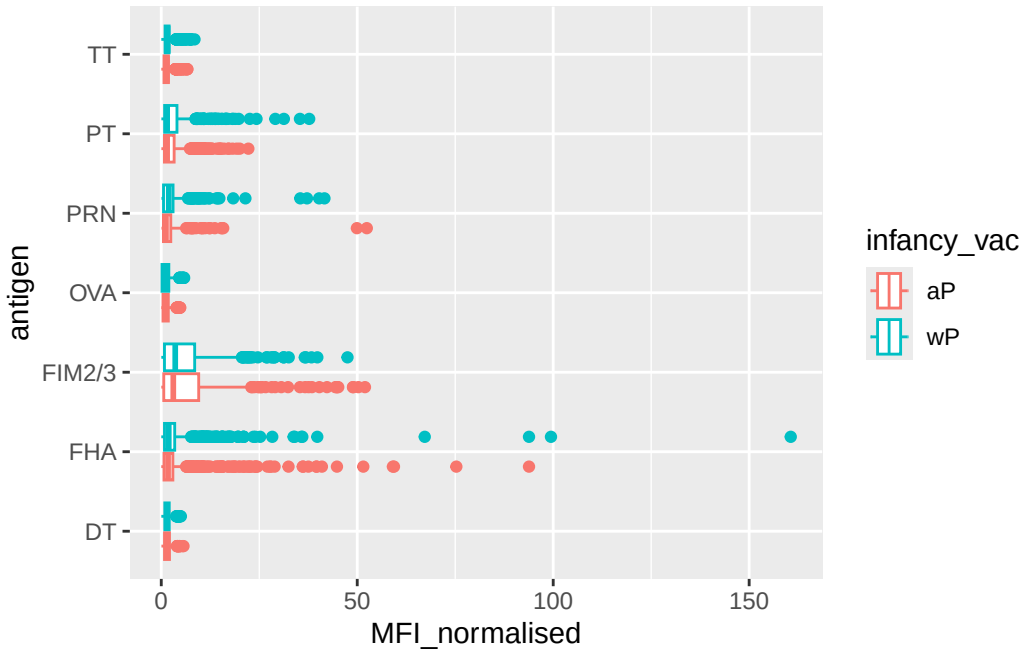
```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



Q. Is there a difference for aP vs wP individuals with these values?

Yes there is a difference between the aP and wP individuals.

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



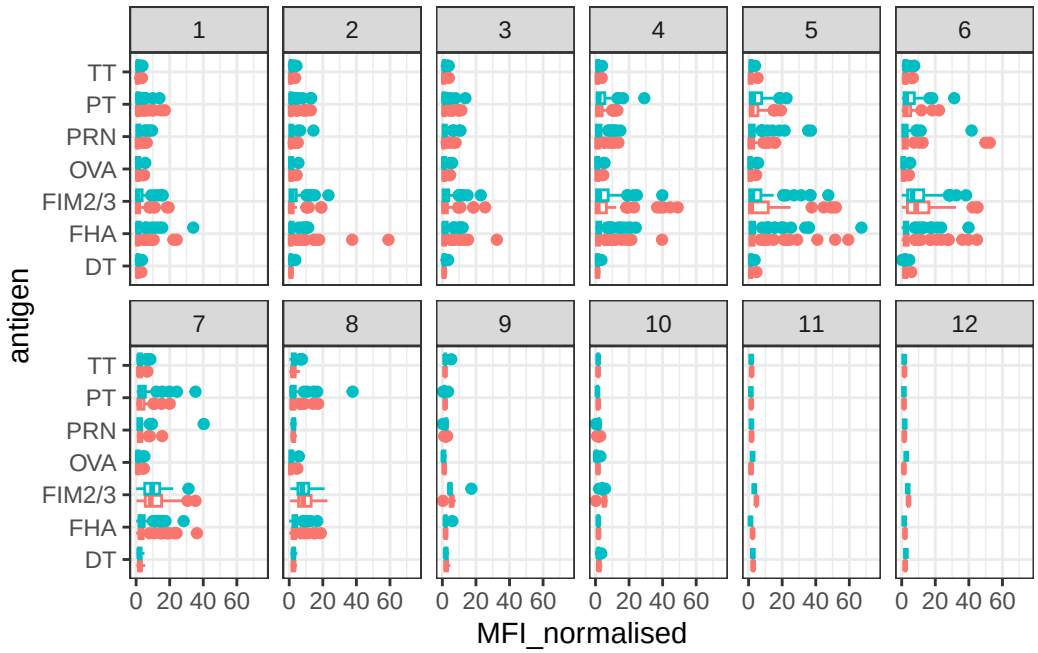
Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens FHA, FIM2/3, PT, and PRN show a large differences in the level of IgG antibody titers recognizing them over time. The levels increase around the time of boost and then they decrease later. These antigens are what makes up of the pertussis vaccine so antibody levels change in response to vaccination. Meanwhile, DT, TT, and OVA show little change over time and this could be due to either unrelated control antigens or represent immunity from prior vaccinations that stays relatively stable.

Q. Is there a temprol response - i.e do values increase or decrease over time?

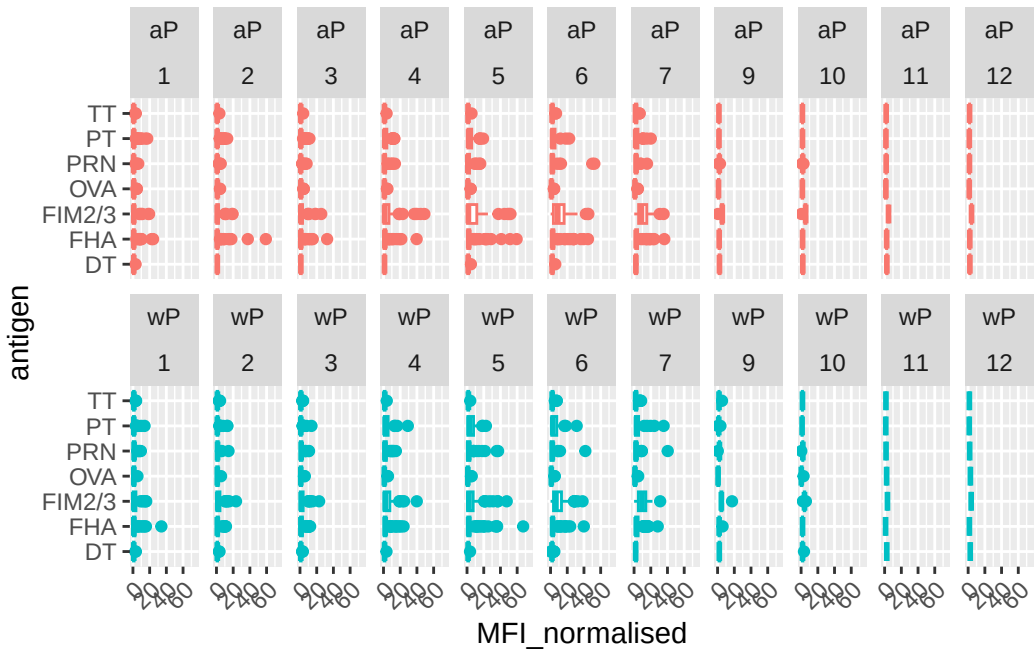
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



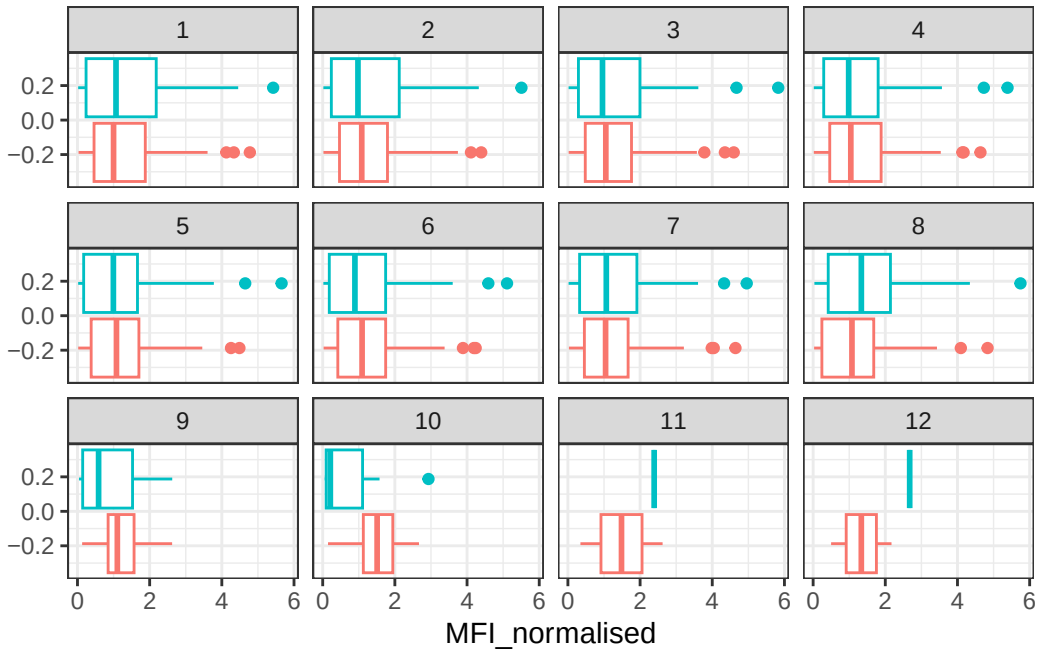

```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2) +
  theme(axis.text.x = element_text(angle = 45, hjust=1))
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

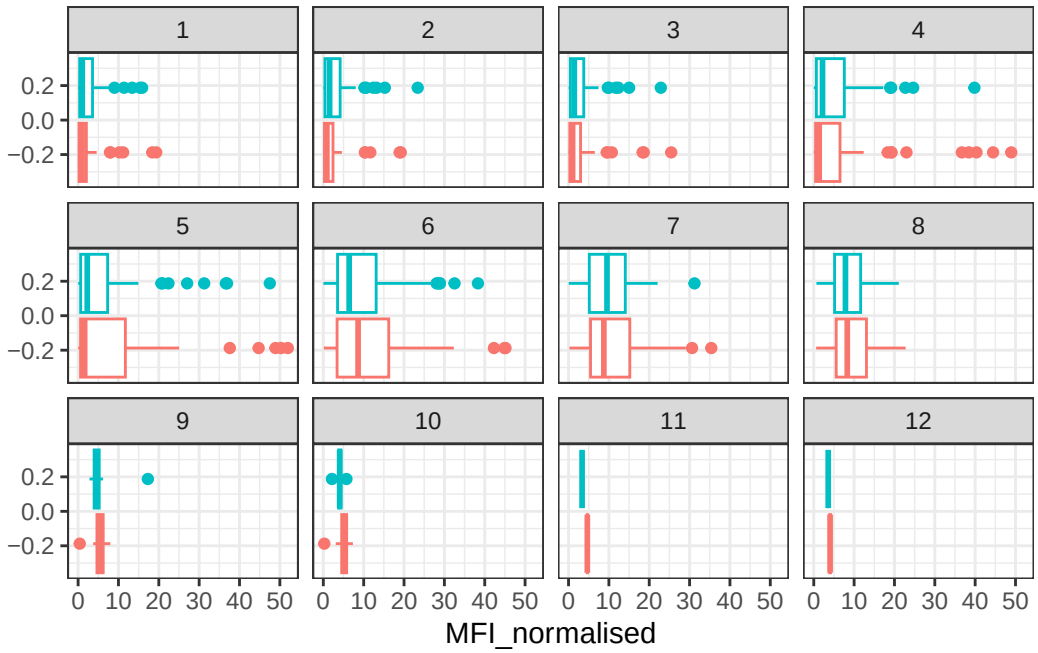


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a "control" antigen ("OVA", that is not in our vaccines) and a clear antigen of interest ("PT", Pertussis Toxin, one of the key virulence factors produced by the bacterium B. pertussis).

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```




```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The PT levels increase over time and it is higher than FIM2/3. It also looks to reach its peak at visit 7 and then it starts decreasing. This trend is similar for wP and aP subjects. The PT antigen shows a dramatic increase in IgG antibody levels after vaccination, especially around visits 4-6. Meanwhile, the FIM2/3 antigen shows lower antibody levels and a more slow change over time. This means that the immune response is specific to pertussis antigens like PT and not to unrelated antigens like FIM2/3.

Q17. Do you see any clear difference in aP vs. wP responses?

Yes, there is a clear difference between aP and wP responses. In many of the visits, individuals who got the wP vaccine show higher IgG antibody levels compared to those who got the aP vaccines. This means that the whole-cell pertussis vaccine may produce a stronger or longer-lasting immune response compared to the acellular pertussis vaccine. The distributions for wP are more towards the right side of the graph compared to aP.

Focus on "PT" Pertusisis Toxin antigen

```r
pt.igg.21 <- igg |> filter (antigen == "PT",
                        dataset  == "2021_dataset")
pt.igg.21
```

```
    specimen_id isotype is_antigen_specific antigen         MFI MFI_normalised
1           468     IgG               FALSE      PT  112.750000     1.00000000
2           469     IgG               FALSE      PT  111.250000     0.98669623
3           470     IgG               FALSE      PT  125.500000     1.11308204
4           471     IgG               FALSE      PT  224.250000     1.98891353
5           472     IgG               FALSE      PT  304.000000     2.69623060
6           473     IgG               FALSE      PT  274.000000     2.43015521
7           474     IgG               FALSE      PT  171.750000     1.52328160
8           475     IgG               FALSE      PT  124.951277     1.10821532
9           476     IgG               FALSE      PT  275.201277     2.44080956
10          477     IgG               FALSE      PT  263.201277     2.33437940
11          478     IgG               FALSE      PT  852.201277     7.55832619
12          479     IgG               FALSE      PT 1548.451277    13.73349248
13          480     IgG               FALSE      PT 1194.451277    10.59380290
14          481     IgG               FALSE      PT  679.951277     6.03061000
15          483     IgG               FALSE      PT  278.701277     2.47185168
16          484     IgG               FALSE      PT  441.451277     3.91531066
17          485     IgG               FALSE      PT  498.451277     4.42085390
18          486     IgG               FALSE      PT 1526.951277    13.54280512
19          487     IgG               FALSE      PT 1916.701277    16.99956787
20          488     IgG               FALSE      PT 1150.701277    10.20577630
21          489     IgG               FALSE      PT  748.951277     6.64258339
22          490     IgG               FALSE      PT  353.201277     3.13260556
23          491     IgG               FALSE      PT  506.701277     4.49402463
24          492     IgG               FALSE      PT  476.701277     4.22794925
25          493     IgG               FALSE      PT  443.201277     3.93083173
26          494     IgG               FALSE      PT  558.451277     4.95300468
27          495     IgG               FALSE      PT  458.451277     4.06608672
28          496     IgG               FALSE      PT  408.451277     3.62262774
29          498     IgG               FALSE      PT   83.750000     0.74279379
30          499     IgG               FALSE      PT   83.750000     0.74279379
31          500     IgG               FALSE      PT  102.000000     0.90465632
32          501     IgG               FALSE      PT  504.500000     4.47450111
33          502     IgG               FALSE      PT 1837.750000    16.29933481
34          503     IgG               FALSE      PT 1243.250000    11.02660754
35          504     IgG               FALSE      PT  780.750000     6.92461197
36          506     IgG               FALSE      PT  169.750000     1.50554324
37          507     IgG               FALSE      PT  147.750000     1.31042129
```

38	508	IgG	FALSE	PT	149.750000	1.32815965
39	509	IgG	FALSE	PT	1849.000000	16.39911308
40	510	IgG	FALSE	PT	2059.500000	18.26607539
41	511	IgG	FALSE	PT	1194.750000	10.59645233
42	512	IgG	FALSE	PT	417.500000	3.70288248
43	513	IgG	FALSE	PT	71.200062	0.63148614
44	514	IgG	FALSE	PT	66.450062	0.58935754
45	515	IgG	FALSE	PT	64.450062	0.57161918
46	516	IgG	FALSE	PT	369.450062	3.27671896
47	517	IgG	FALSE	PT	2128.450062	18.87760587
48	518	IgG	FALSE	PT	1606.200062	14.24567683
49	519	IgG	FALSE	PT	821.450062	7.28558814
50	521	IgG	FALSE	PT	121.447441	1.07713917
51	522	IgG	FALSE	PT	120.197441	1.06605269
52	523	IgG	FALSE	PT	124.197441	1.10152941
53	524	IgG	FALSE	PT	939.697441	8.33434538
54	525	IgG	FALSE	PT	1565.447441	13.88423451
55	526	IgG	FALSE	PT	1086.947441	9.64033207
56	527	IgG	FALSE	PT	535.947441	4.75341411
57	529	IgG	FALSE	PT	124.450062	1.10376995
58	530	IgG	FALSE	PT	125.950062	1.11707372
59	531	IgG	FALSE	PT	121.950062	1.08159701
60	532	IgG	FALSE	PT	719.450062	6.38093182
61	533	IgG	FALSE	PT	1150.200062	10.20133093
62	534	IgG	FALSE	PT	782.700062	6.94190743
63	535	IgG	FALSE	PT	395.200062	3.50510033
64	537	IgG	FALSE	PT	43.451277	0.38537718
65	538	IgG	FALSE	PT	64.701277	0.57384725
66	539	IgG	FALSE	PT	56.701277	0.50289381
67	540	IgG	FALSE	PT	50.701277	0.44967874
68	541	IgG	FALSE	PT	201.201277	1.78449027
69	542	IgG	FALSE	PT	121.451277	1.07717319
70	543	IgG	FALSE	PT	156.701277	1.38981177
71	546	IgG	FALSE	PT	732.701277	6.49845922
72	547	IgG	FALSE	PT	959.451277	8.50954570
73	548	IgG	FALSE	PT	740.201277	6.56497807
74	549	IgG	FALSE	PT	723.201277	6.41420202
75	550	IgG	FALSE	PT	916.951277	8.13260556
76	551	IgG	FALSE	PT	637.201277	5.65145257
77	552	IgG	FALSE	PT	631.201277	5.59823749
78	554	IgG	FALSE	PT	366.750000	3.25277162
79	555	IgG	FALSE	PT	285.500000	2.53215078
80	556	IgG	FALSE	PT	359.500000	3.18847007

81	557	IgG	FALSE	PT	963.750000	8.54767184
82	558	IgG	FALSE	PT	1736.250000	15.39911308
83	559	IgG	FALSE	PT	1120.750000	9.94013304
84	560	IgG	FALSE	PT	610.000000	5.41019956
85	562	IgG	FALSE	PT	303.197441	2.68911256
86	563	IgG	FALSE	PT	325.197441	2.88423451
87	564	IgG	FALSE	PT	344.947441	3.05940081
88	565	IgG	FALSE	PT	981.947441	8.70906821
89	566	IgG	FALSE	PT	1349.697441	11.97070901
90	567	IgG	FALSE	PT	857.947441	7.60928994
91	568	IgG	FALSE	PT	640.947441	5.68467797
92	569	IgG	FALSE	PT	48.947441	0.43412365
93	570	IgG	FALSE	PT	53.947441	0.47846954
94	571	IgG	FALSE	PT	58.947441	0.52281544
95	572	IgG	FALSE	PT	250.947441	2.22569793
96	573	IgG	FALSE	PT	385.697441	3.42081988
97	574	IgG	FALSE	PT	227.947441	2.02170679
98	575	IgG	FALSE	PT	141.447441	1.25452276
99	577	IgG	FALSE	PT	14.750000	0.13082040
100	578	IgG	FALSE	PT	9.750000	0.08647450
101	579	IgG	FALSE	PT	11.750000	0.10421286
102	580	IgG	FALSE	PT	52.250000	0.46341463
103	581	IgG	FALSE	PT	136.250000	1.20842572
104	582	IgG	FALSE	PT	120.500000	1.06873614
105	583	IgG	FALSE	PT	65.750000	0.58314856
106	585	IgG	FALSE	PT	1195.500000	10.60310421
107	586	IgG	FALSE	PT	1105.750000	9.80709534
108	587	IgG	FALSE	PT	1124.250000	9.97117517
109	588	IgG	FALSE	PT	1129.750000	10.01995565
110	589	IgG	FALSE	PT	1247.000000	11.05986696
111	590	IgG	FALSE	PT	1028.750000	9.12416851
112	591	IgG	FALSE	PT	582.750000	5.16851441
113	593	IgG	FALSE	PT	69.250000	0.61419069
114	594	IgG	FALSE	PT	91.250000	0.80931264
115	595	IgG	FALSE	PT	62.250000	0.55210643
116	596	IgG	FALSE	PT	224.000000	1.98669623
117	597	IgG	FALSE	PT	506.500000	4.49223947
118	598	IgG	FALSE	PT	307.250000	2.72505543
119	599	IgG	FALSE	PT	248.750000	2.20620843
120	601	IgG	FALSE	PT	112.000000	0.99334812
121	602	IgG	FALSE	PT	122.000000	1.08203991
122	603	IgG	FALSE	PT	106.250000	0.94235033
123	604	IgG	FALSE	PT	495.000000	4.39024390

124	605	IgG	FALSE	PT	831.500000	7.37472284
125	606	IgG	FALSE	PT	550.500000	4.88248337
126	607	IgG	FALSE	PT	371.000000	3.29046563
127	608	IgG	FALSE	PT	451.250000	4.00221729
128	609	IgG	FALSE	PT	390.750000	3.46563193
129	610	IgG	FALSE	PT	446.250000	3.95787140
130	611	IgG	FALSE	PT	489.750000	4.34368071
131	612	IgG	FALSE	PT	1019.250000	9.03991131
132	613	IgG	FALSE	PT	1023.250000	9.07538803
133	614	IgG	FALSE	PT	874.250000	7.75388027
134	616	IgG	FALSE	PT	43.250000	0.38359202
135	617	IgG	FALSE	PT	46.750000	0.41463415
136	618	IgG	FALSE	PT	41.250000	0.36585366
137	619	IgG	FALSE	PT	133.500000	1.18403548
138	620	IgG	FALSE	PT	248.500000	2.20399113
139	621	IgG	FALSE	PT	291.500000	2.58536585
140	622	IgG	FALSE	PT	208.250000	1.84700665
141	623	IgG	FALSE	PT	284.250000	2.52106430
142	624	IgG	FALSE	PT	291.250000	2.58314856
143	625	IgG	FALSE	PT	299.500000	2.65631929
144	626	IgG	FALSE	PT	331.500000	2.94013304
145	627	IgG	FALSE	PT	516.750000	4.58314856
146	628	IgG	FALSE	PT	477.750000	4.23725055
147	629	IgG	FALSE	PT	414.000000	3.67184035
148	636	IgG	FALSE	PT	157.950062	1.40088747
149	637	IgG	FALSE	PT	147.200062	1.30554379
150	638	IgG	FALSE	PT	152.700062	1.35432428
151	639	IgG	FALSE	PT	331.200062	2.93747284
152	640	IgG	FALSE	PT	436.450062	3.87095399
153	641	IgG	FALSE	PT	381.950062	3.38758370
154	642	IgG	FALSE	PT	213.200062	1.89090964
155	643	IgG	FALSE	PT	24.201277	0.21464548
156	644	IgG	FALSE	PT	59.701277	0.52950135
157	645	IgG	FALSE	PT	56.701277	0.50289381
158	646	IgG	FALSE	PT	152.701277	1.35433505
159	647	IgG	FALSE	PT	474.951277	4.21242818
160	648	IgG	FALSE	PT	289.201277	2.56497807
161	649	IgG	FALSE	PT	217.701277	1.93083173
162	650	IgG	FALSE	PT	29.250000	0.25942350
163	651	IgG	FALSE	PT	32.750000	0.29046563
164	652	IgG	FALSE	PT	32.250000	0.28603104
165	653	IgG	FALSE	PT	107.750000	0.95565410
166	654	IgG	FALSE	PT	168.750000	1.49667406

167	655	IgG	FALSE	PT	492.750000	4.37028825
168	656	IgG	FALSE	PT	42.250000	0.37472284
169	657	IgG	FALSE	PT	26.450062	0.23459035
170	658	IgG	FALSE	PT	23.450062	0.20798281
171	659	IgG	FALSE	PT	373.700062	3.31441297
172	660	IgG	FALSE	PT	471.700062	4.18359257
173	661	IgG	FALSE	PT	615.200062	5.45631984
174	662	IgG	FALSE	PT	337.200062	2.99068791
175	663	IgG	FALSE	PT	116.200062	1.03059922
176	674	IgG	FALSE	PT	5.447441	0.04831433
177	675	IgG	FALSE	PT	6.447441	0.05718351
178	676	IgG	FALSE	PT	5.197441	0.04609704
179	677	IgG	FALSE	PT	67.197441	0.59598617
180	678	IgG	FALSE	PT	133.947441	1.18800391
181	679	IgG	FALSE	PT	82.947441	0.73567575
182	680	IgG	FALSE	PT	50.947441	0.45186201
183	681	IgG	FALSE	PT	29.950062	0.26563248
184	682	IgG	FALSE	PT	27.950062	0.24789412
185	683	IgG	FALSE	PT	30.450062	0.27006707
186	684	IgG	FALSE	PT	76.950062	0.68248392
187	685	IgG	FALSE	PT	221.950062	1.96851497
188	686	IgG	FALSE	PT	193.700062	1.71796064
189	687	IgG	FALSE	PT	67.950062	0.60266131
190	688	IgG	FALSE	PT	101.697441	0.90197287
191	689	IgG	FALSE	PT	83.947441	0.74454493
192	690	IgG	FALSE	PT	106.697441	0.94631877
193	691	IgG	FALSE	PT	270.447441	2.39864693
194	692	IgG	FALSE	PT	487.447441	4.32325890
195	693	IgG	FALSE	PT	385.947441	3.42303717
196	694	IgG	FALSE	PT	252.697441	2.24121899
197	695	IgG	FALSE	PT	20.250000	0.17960089
198	696	IgG	FALSE	PT	22.250000	0.19733925
199	697	IgG	FALSE	PT	25.250000	0.22394678
200	698	IgG	FALSE	PT	180.250000	1.59866962
201	699	IgG	FALSE	PT	377.750000	3.35033259
202	700	IgG	FALSE	PT	323.500000	2.86917960
203	701	IgG	FALSE	PT	182.250000	1.61640798
204	702	IgG	FALSE	PT	223.750000	1.98447894
205	703	IgG	FALSE	PT	215.750000	1.91352550
206	704	IgG	FALSE	PT	220.750000	1.95787140
207	705	IgG	FALSE	PT	586.250000	5.19955654
208	706	IgG	FALSE	PT	645.250000	5.72283814
209	707	IgG	FALSE	PT	596.250000	5.28824834

210	708	IgG	FALSE	PT	396.250000	3.51441242
211	709	IgG	FALSE	PT	599.000000	5.31263858
212	710	IgG	FALSE	PT	557.250000	4.94235033
213	711	IgG	FALSE	PT	519.250000	4.60532151
214	712	IgG	FALSE	PT	709.750000	6.29490022
215	713	IgG	FALSE	PT	993.750000	8.81374723
216	714	IgG	FALSE	PT	877.000000	7.77827051
217	715	IgG	FALSE	PT	639.250000	5.66962306
218	716	IgG	FALSE	PT	441.697441	3.91749393
219	717	IgG	FALSE	PT	494.197441	4.38312586
220	718	IgG	FALSE	PT	484.947441	4.30108595
221	719	IgG	FALSE	PT	661.947441	5.87093074
222	720	IgG	FALSE	PT	1078.697441	9.56716134
223	721	IgG	FALSE	PT	856.697441	7.59820347
224	722	IgG	FALSE	PT	651.947441	5.78223895
225	723	IgG	FALSE	PT	78.750000	0.69844789
226	724	IgG	FALSE	PT	87.750000	0.77827051
227	725	IgG	FALSE	PT	78.750000	0.69844789
228	726	IgG	FALSE	PT	140.250000	1.24390244
229	727	IgG	FALSE	PT	386.000000	3.42350333
230	728	IgG	FALSE	PT	331.000000	2.93569845
231	729	IgG	FALSE	PT	304.750000	2.70288248

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	MFI	5.197441	61	-4
2	MFI	5.197441	61	1
3	MFI	5.197441	61	3
4	MFI	5.197441	61	7
5	MFI	5.197441	61	14
6	MFI	5.197441	61	30
7	MFI	5.197441	61	91
8	MFI	5.197441	62	0
9	MFI	5.197441	62	1
10	MFI	5.197441	62	3
11	MFI	5.197441	62	7
12	MFI	5.197441	62	14
13	MFI	5.197441	62	30
14	MFI	5.197441	62	101
15	MFI	5.197441	63	0
16	MFI	5.197441	63	1
17	MFI	5.197441	63	3
18	MFI	5.197441	63	7
19	MFI	5.197441	63	14
20	MFI	5.197441	63	38

21	MFI	5.197441	63	121
22	MFI	5.197441	64	0
23	MFI	5.197441	64	1
24	MFI	5.197441	64	3
25	MFI	5.197441	64	7
26	MFI	5.197441	64	14
27	MFI	5.197441	64	30
28	MFI	5.197441	64	101
29	MFI	5.197441	65	0
30	MFI	5.197441	65	1
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3	3	Blood	3	wP

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5	14	Blood	5	wP
6	30	Blood	6	wP
7	120	Blood	7	wP
8	0	Blood	1	wP
9	1	Blood	2	wP
10	3	Blood	3	wP
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41	30	Blood	6	wP
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45	3	Blood	3	wP
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69	30	Blood	6	aP
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214	7	Blood	4	aP
215	14	Blood	5	aP
216	30	Blood	6	aP
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199	Female	Hispanic or Latino
200	Female	Hispanic or Latino
201	Female	Hispanic or Latino

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1                     Unknown or Not Reported    1987-01-01    2019-04-08
2                     Unknown or Not Reported    1987-01-01    2019-04-08
3                     Unknown or Not Reported    1987-01-01    2019-04-08
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5                     Unknown or Not Reported    1987-01-01    2019-04-08
6                     Unknown or Not Reported    1987-01-01    2019-04-08
7                     Unknown or Not Reported    1987-01-01    2019-04-08
8                                       Asian    1993-01-01    2018-11-26
9                                       Asian    1993-01-01    2018-11-26
10                                      Asian    1993-01-01    2018-11-26
11                                      Asian    1993-01-01    2018-11-26
12                                      Asian    1993-01-01    2018-11-26
```

13	Asian	1993-01-01	2018-11-26
14	Asian	1993-01-01	2018-11-26
15	White	1995-01-01	2018-11-26
16	White	1995-01-01	2018-11-26
17	White	1995-01-01	2018-11-26
18	White	1995-01-01	2018-11-26
19	White	1995-01-01	2018-11-26
20	White	1995-01-01	2018-11-26
21	White	1995-01-01	2018-11-26
22	Asian	1993-01-01	2018-11-26
23	Asian	1993-01-01	2018-11-26
24	Asian	1993-01-01	2018-11-26
25	Asian	1993-01-01	2018-11-26
26	Asian	1993-01-01	2018-11-26
27	Asian	1993-01-01	2018-11-26
28	Asian	1993-01-01	2018-11-26
29	White	1990-01-01	2018-12-03
30	White	1990-01-01	2018-12-03
31	White	1990-01-01	2018-12-03
32	White	1990-01-01	2018-12-03
33	White	1990-01-01	2018-12-03
34	White	1990-01-01	2018-12-03
35	White	1990-01-01	2018-12-03
36	Black or African American	1976-01-01	2018-12-03
37	Black or African American	1976-01-01	2018-12-03
38	Black or African American	1976-01-01	2018-12-03
39	Black or African American	1976-01-01	2018-12-03
40	Black or African American	1976-01-01	2018-12-03
41	Black or African American	1976-01-01	2018-12-03
42	Black or African American	1976-01-01	2018-12-03
43	White	1972-01-01	2019-01-28
44	White	1972-01-01	2019-01-28
45	White	1972-01-01	2019-01-28
46	White	1972-01-01	2019-01-28
47	White	1972-01-01	2019-01-28
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49	White	1972-01-01	2019-01-28
50	White	1972-01-01	2019-01-28
51	White	1972-01-01	2019-01-28
52	White	1972-01-01	2019-01-28
53	White	1972-01-01	2019-01-28
54	White	1972-01-01	2019-01-28
55	White	1972-01-01	2019-01-28

56	White	1972-01-01	2019-01-28
57	White	1990-01-01	2019-01-28
58	White	1990-01-01	2019-01-28
59	White	1990-01-01	2019-01-28
60	White	1990-01-01	2019-01-28
61	White	1990-01-01	2019-01-28
62	White	1990-01-01	2019-01-28
63	White	1990-01-01	2019-01-28
64	American Indian/Alaska Native	1998-01-01	2019-01-28
65	American Indian/Alaska Native	1998-01-01	2019-01-28
66	American Indian/Alaska Native	1998-01-01	2019-01-28
67	American Indian/Alaska Native	1998-01-01	2019-01-28
68	American Indian/Alaska Native	1998-01-01	2019-01-28
69	American Indian/Alaska Native	1998-01-01	2019-01-28
70	American Indian/Alaska Native	1998-01-01	2019-01-28
71	White	1998-01-01	2019-01-28
72	White	1998-01-01	2019-01-28
73	White	1998-01-01	2019-01-28
74	White	1998-01-01	2019-01-28
75	White	1998-01-01	2019-01-28
76	White	1998-01-01	2019-01-28
77	White	1998-01-01	2019-01-28
78	White	1991-01-01	2019-02-25
79	White	1991-01-01	2019-02-25
80	White	1991-01-01	2019-02-25
81	White	1991-01-01	2019-02-25
82	White	1991-01-01	2019-02-25
83	White	1991-01-01	2019-02-25
84	White	1991-01-01	2019-02-25
85	White	1995-01-01	2019-02-25
86	White	1995-01-01	2019-02-25
87	White	1995-01-01	2019-02-25
88	White	1995-01-01	2019-02-25
89	White	1995-01-01	2019-02-25
90	White	1995-01-01	2019-02-25
91	White	1995-01-01	2019-02-25
92	White	1995-01-01	2019-02-25
93	White	1995-01-01	2019-02-25
94	White	1995-01-01	2019-02-25
95	White	1995-01-01	2019-02-25
96	White	1995-01-01	2019-02-25
97	White	1995-01-01	2019-02-25
98	White	1995-01-01	2019-02-25

99	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
100	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
101	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
102	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
103	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
104	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
105	Native Hawaiian or Other Pacific Islander	1998-01-01	2019-02-25
106	Asian	1998-01-01	2019-02-25
107	Asian	1998-01-01	2019-02-25
108	Asian	1998-01-01	2019-02-25
109	Asian	1998-01-01	2019-02-25
110	Asian	1998-01-01	2019-02-25
111	Asian	1998-01-01	2019-02-25
112	Asian	1998-01-01	2019-02-25
113	White	1988-01-01	2019-03-18
114	White	1988-01-01	2019-03-18
115	White	1988-01-01	2019-03-18
116	White	1988-01-01	2019-03-18
117	White	1988-01-01	2019-03-18
118	White	1988-01-01	2019-03-18
119	White	1988-01-01	2019-03-18
120	White	1993-01-01	2019-03-18
121	White	1993-01-01	2019-03-18
122	White	1993-01-01	2019-03-18
123	White	1993-01-01	2019-03-18
124	White	1993-01-01	2019-03-18
125	White	1993-01-01	2019-03-18
126	White	1993-01-01	2019-03-18
127	White	1987-01-01	2019-03-18
128	White	1987-01-01	2019-03-18
129	White	1987-01-01	2019-03-18
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133	White	1987-01-01	2019-03-18
134	Asian	1992-01-01	2019-03-18
135	Asian	1992-01-01	2019-03-18
136	Asian	1992-01-01	2019-03-18
137	Asian	1992-01-01	2019-03-18
138	Asian	1992-01-01	2019-03-18
139	Asian	1992-01-01	2019-03-18
140	Asian	1992-01-01	2019-03-18
141	White	1993-01-01	2019-03-18

142	White	1993-01-01	2019-03-18
143	White	1993-01-01	2019-03-18
144	White	1993-01-01	2019-03-18
145	White	1993-01-01	2019-03-18
146	White	1993-01-01	2019-03-18
147	White	1993-01-01	2019-03-18
148	White	1999-01-01	2019-04-08
149	White	1999-01-01	2019-04-08
150	White	1999-01-01	2019-04-08
151	White	1999-01-01	2019-04-08
152	White	1999-01-01	2019-04-08
153	White	1999-01-01	2019-04-08
154	White	1999-01-01	2019-04-08
155	More Than One Race	1997-01-01	2019-04-08
156	More Than One Race	1997-01-01	2019-04-08
157	More Than One Race	1997-01-01	2019-04-08
158	More Than One Race	1997-01-01	2019-04-08
159	More Than One Race	1997-01-01	2019-04-08
160	More Than One Race	1997-01-01	2019-04-08
161	More Than One Race	1997-01-01	2019-04-08
162	White	2000-01-01	2019-04-29
163	White	2000-01-01	2019-04-29
164	White	2000-01-01	2019-04-29
165	White	2000-01-01	2019-04-29
166	White	2000-01-01	2019-04-29
167	White	2000-01-01	2019-04-29
168	White	2000-01-01	2019-04-29
169	Asian	1998-01-01	2019-04-29
170	Asian	1998-01-01	2019-04-29
171	Asian	1998-01-01	2019-04-29
172	Asian	1998-01-01	2019-04-29
173	Asian	1998-01-01	2019-04-29
174	Asian	1998-01-01	2019-04-29
175	Asian	1998-01-01	2019-04-29
176	Asian	1997-01-01	2019-06-03
177	Asian	1997-01-01	2019-06-03
178	Asian	1997-01-01	2019-06-03
179	Asian	1997-01-01	2019-06-03
180	Asian	1997-01-01	2019-06-03
181	Asian	1997-01-01	2019-06-03
182	Asian	1997-01-01	2019-06-03
183	Asian	1999-01-01	2019-06-03
184	Asian	1999-01-01	2019-06-03

185	Asian	1999-01-01	2019-06-03
186	Asian	1999-01-01	2019-06-03
187	Asian	1999-01-01	2019-06-03
188	Asian	1999-01-01	2019-06-03
189	Asian	1999-01-01	2019-06-03
190	Unknown or Not Reported	1998-01-01	2019-06-03
191	Unknown or Not Reported	1998-01-01	2019-06-03
192	Unknown or Not Reported	1998-01-01	2019-06-03
193	Unknown or Not Reported	1998-01-01	2019-06-03
194	Unknown or Not Reported	1998-01-01	2019-06-03
195	Unknown or Not Reported	1998-01-01	2019-06-03
196	Unknown or Not Reported	1998-01-01	2019-06-03
197	White	2000-01-01	2019-06-24
198	White	2000-01-01	2019-06-24
199	White	2000-01-01	2019-06-24
200	White	2000-01-01	2019-06-24
201	White	2000-01-01	2019-06-24
202	White	2000-01-01	2019-06-24
203	White	2000-01-01	2019-06-24
204	More Than One Race	1996-01-01	2019-06-24
205	More Than One Race	1996-01-01	2019-06-24
206	More Than One Race	1996-01-01	2019-06-24
207	More Than One Race	1996-01-01	2019-06-24
208	More Than One Race	1996-01-01	2019-06-24
209	More Than One Race	1996-01-01	2019-06-24
210	More Than One Race	1996-01-01	2019-06-24
211	Unknown or Not Reported	1999-01-01	2019-06-24
212	Unknown or Not Reported	1999-01-01	2019-06-24
213	Unknown or Not Reported	1999-01-01	2019-06-24
214	Unknown or Not Reported	1999-01-01	2019-06-24
215	Unknown or Not Reported	1999-01-01	2019-06-24
216	Unknown or Not Reported	1999-01-01	2019-06-24
217	Unknown or Not Reported	1999-01-01	2019-06-24
218	Unknown or Not Reported	1998-01-01	2019-06-24
219	Unknown or Not Reported	1998-01-01	2019-06-24
220	Unknown or Not Reported	1998-01-01	2019-06-24
221	Unknown or Not Reported	1998-01-01	2019-06-24
222	Unknown or Not Reported	1998-01-01	2019-06-24
223	Unknown or Not Reported	1998-01-01	2019-06-24
224	Unknown or Not Reported	1998-01-01	2019-06-24
225	Unknown or Not Reported	2000-01-01	2019-06-24
226	Unknown or Not Reported	2000-01-01	2019-06-24
227	Unknown or Not Reported	2000-01-01	2019-06-24

```
228                    Unknown or Not Reported    2000-01-01    2019-06-24
229                    Unknown or Not Reported    2000-01-01    2019-06-24
230                    Unknown or Not Reported    2000-01-01    2019-06-24
231                    Unknown or Not Reported    2000-01-01    2019-06-24
         dataset        age
1   2021_dataset 14311 days
2   2021_dataset 14311 days
3   2021_dataset 14311 days
4   2021_dataset 14311 days
5   2021_dataset 14311 days
6   2021_dataset 14311 days
7   2021_dataset 14311 days
8   2021_dataset 12119 days
9   2021_dataset 12119 days
10  2021_dataset 12119 days
11  2021_dataset 12119 days
12  2021_dataset 12119 days
13  2021_dataset 12119 days
14  2021_dataset 12119 days
15  2021_dataset 11389 days
16  2021_dataset 11389 days
17  2021_dataset 11389 days
18  2021_dataset 11389 days
19  2021_dataset 11389 days
20  2021_dataset 11389 days
21  2021_dataset 11389 days
22  2021_dataset 12119 days
23  2021_dataset 12119 days
24  2021_dataset 12119 days
25  2021_dataset 12119 days
26  2021_dataset 12119 days
27  2021_dataset 12119 days
28  2021_dataset 12119 days
29  2021_dataset 13215 days
30  2021_dataset 13215 days
31  2021_dataset 13215 days
32  2021_dataset 13215 days
33  2021_dataset 13215 days
34  2021_dataset 13215 days
35  2021_dataset 13215 days
36  2021_dataset 18329 days
37  2021_dataset 18329 days
38  2021_dataset 18329 days
```

```
39  2021_dataset 18329 days
40  2021_dataset 18329 days
41  2021_dataset 18329 days
42  2021_dataset 18329 days
43  2021_dataset 19790 days
44  2021_dataset 19790 days
45  2021_dataset 19790 days
46  2021_dataset 19790 days
47  2021_dataset 19790 days
48  2021_dataset 19790 days
49  2021_dataset 19790 days
50  2021_dataset 19790 days
51  2021_dataset 19790 days
52  2021_dataset 19790 days
53  2021_dataset 19790 days
54  2021_dataset 19790 days
55  2021_dataset 19790 days
56  2021_dataset 19790 days
57  2021_dataset 13215 days
58  2021_dataset 13215 days
59  2021_dataset 13215 days
60  2021_dataset 13215 days
61  2021_dataset 13215 days
62  2021_dataset 13215 days
63  2021_dataset 13215 days
64  2021_dataset 10293 days
65  2021_dataset 10293 days
66  2021_dataset 10293 days
67  2021_dataset 10293 days
68  2021_dataset 10293 days
69  2021_dataset 10293 days
70  2021_dataset 10293 days
71  2021_dataset 10293 days
72  2021_dataset 10293 days
73  2021_dataset 10293 days
74  2021_dataset 10293 days
75  2021_dataset 10293 days
76  2021_dataset 10293 days
77  2021_dataset 10293 days
78  2021_dataset 12850 days
79  2021_dataset 12850 days
80  2021_dataset 12850 days
81  2021_dataset 12850 days
```

```
82  2021_dataset 12850 days
83  2021_dataset 12850 days
84  2021_dataset 12850 days
85  2021_dataset 11389 days
86  2021_dataset 11389 days
87  2021_dataset 11389 days
88  2021_dataset 11389 days
89  2021_dataset 11389 days
90  2021_dataset 11389 days
91  2021_dataset 11389 days
92  2021_dataset 11389 days
93  2021_dataset 11389 days
94  2021_dataset 11389 days
95  2021_dataset 11389 days
96  2021_dataset 11389 days
97  2021_dataset 11389 days
98  2021_dataset 11389 days
99  2021_dataset 10293 days
100 2021_dataset 10293 days
101 2021_dataset 10293 days
102 2021_dataset 10293 days
103 2021_dataset 10293 days
104 2021_dataset 10293 days
105 2021_dataset 10293 days
106 2021_dataset 10293 days
107 2021_dataset 10293 days
108 2021_dataset 10293 days
109 2021_dataset 10293 days
110 2021_dataset 10293 days
111 2021_dataset 10293 days
112 2021_dataset 10293 days
113 2021_dataset 13946 days
114 2021_dataset 13946 days
115 2021_dataset 13946 days
116 2021_dataset 13946 days
117 2021_dataset 13946 days
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119 2021_dataset 13946 days
120 2021_dataset 12119 days
121 2021_dataset 12119 days
122 2021_dataset 12119 days
123 2021_dataset 12119 days
124 2021_dataset 12119 days
```

```
125 2021_dataset 12119 days
126 2021_dataset 12119 days
127 2021_dataset 14311 days
128 2021_dataset 14311 days
129 2021_dataset 14311 days
130 2021_dataset 14311 days
131 2021_dataset 14311 days
132 2021_dataset 14311 days
133 2021_dataset 14311 days
134 2021_dataset 12485 days
135 2021_dataset 12485 days
136 2021_dataset 12485 days
137 2021_dataset 12485 days
138 2021_dataset 12485 days
139 2021_dataset 12485 days
140 2021_dataset 12485 days
141 2021_dataset 12119 days
142 2021_dataset 12119 days
143 2021_dataset 12119 days
144 2021_dataset 12119 days
145 2021_dataset 12119 days
146 2021_dataset 12119 days
147 2021_dataset 12119 days
148 2021_dataset  9928 days
149 2021_dataset  9928 days
150 2021_dataset  9928 days
151 2021_dataset  9928 days
152 2021_dataset  9928 days
153 2021_dataset  9928 days
154 2021_dataset  9928 days
155 2021_dataset 10658 days
156 2021_dataset 10658 days
157 2021_dataset 10658 days
158 2021_dataset 10658 days
159 2021_dataset 10658 days
160 2021_dataset 10658 days
161 2021_dataset 10658 days
162 2021_dataset  9563 days
163 2021_dataset  9563 days
164 2021_dataset  9563 days
165 2021_dataset  9563 days
166 2021_dataset  9563 days
167 2021_dataset  9563 days
```

```
168 2021_dataset  9563 days
169 2021_dataset 10293 days
170 2021_dataset 10293 days
171 2021_dataset 10293 days
172 2021_dataset 10293 days
173 2021_dataset 10293 days
174 2021_dataset 10293 days
175 2021_dataset 10293 days
176 2021_dataset 10658 days
177 2021_dataset 10658 days
178 2021_dataset 10658 days
179 2021_dataset 10658 days
180 2021_dataset 10658 days
181 2021_dataset 10658 days
182 2021_dataset 10658 days
183 2021_dataset  9928 days
184 2021_dataset  9928 days
185 2021_dataset  9928 days
186 2021_dataset  9928 days
187 2021_dataset  9928 days
188 2021_dataset  9928 days
189 2021_dataset  9928 days
190 2021_dataset 10293 days
191 2021_dataset 10293 days
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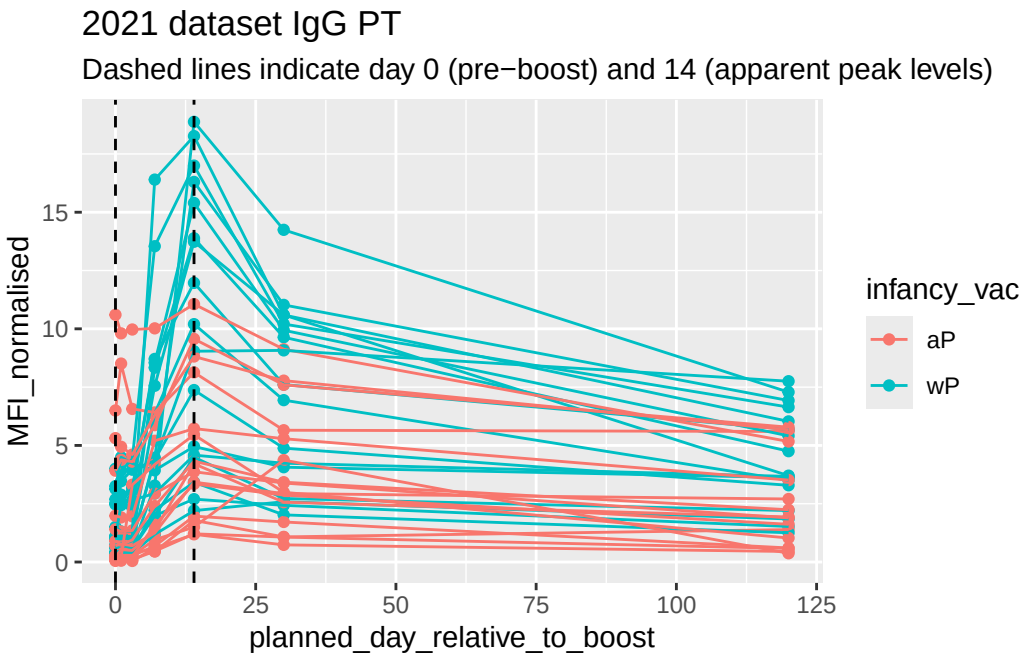
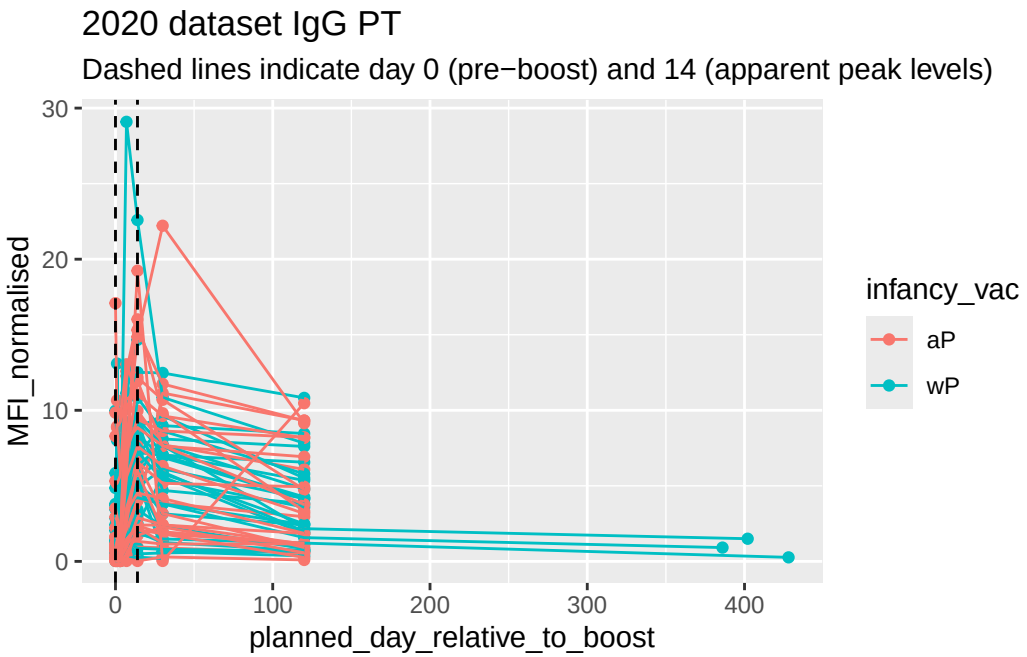
193 2021_dataset 10293 days
194 2021_dataset 10293 days
195 2021_dataset 10293 days
196 2021_dataset 10293 days
197 2021_dataset  9563 days
198 2021_dataset  9563 days
199 2021_dataset  9563 days
200 2021_dataset  9563 days
201 2021_dataset  9563 days
202 2021_dataset  9563 days
203 2021_dataset  9563 days
204 2021_dataset 11024 days
205 2021_dataset 11024 days
206 2021_dataset 11024 days
207 2021_dataset 11024 days
208 2021_dataset 11024 days
209 2021_dataset 11024 days
210 2021_dataset 11024 days
```

```
211 2021_dataset  9928 days
212 2021_dataset  9928 days
213 2021_dataset  9928 days
214 2021_dataset  9928 days
215 2021_dataset  9928 days
216 2021_dataset  9928 days
217 2021_dataset  9928 days
218 2021_dataset 10293 days
219 2021_dataset 10293 days
220 2021_dataset 10293 days
221 2021_dataset 10293 days
222 2021_dataset 10293 days
223 2021_dataset 10293 days
224 2021_dataset 10293 days
225 2021_dataset  9563 days
226 2021_dataset  9563 days
227 2021_dataset  9563 days
228 2021_dataset  9563 days
229 2021_dataset  9563 days
230 2021_dataset  9563 days
231 2021_dataset  9563 days
```

```r
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG",  antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



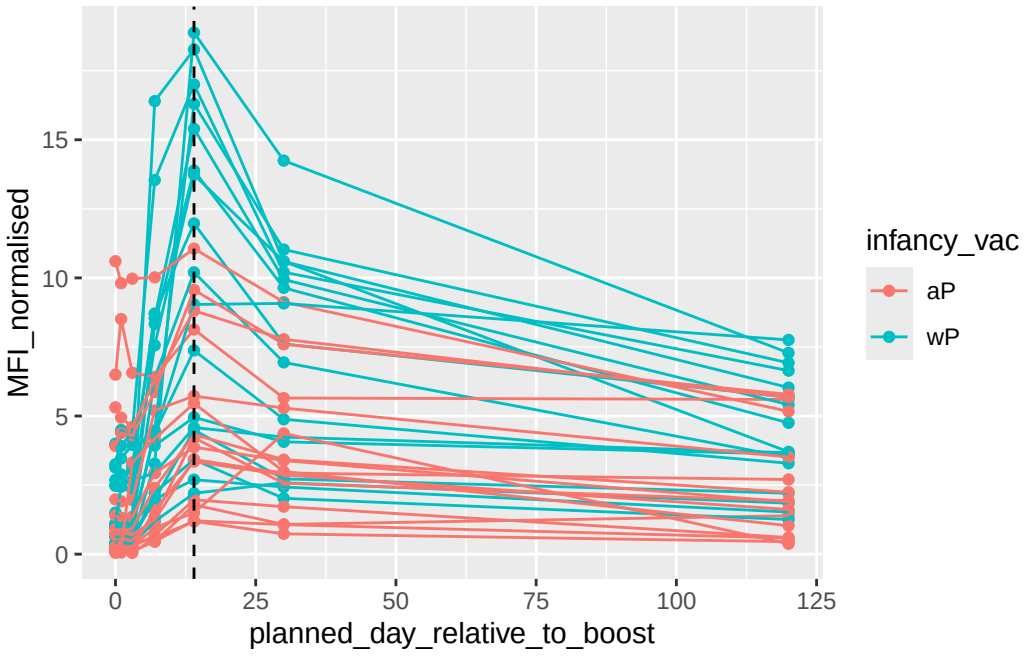

```
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG",  antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2020 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Q18. Does this trend look similar for the 2020 dataset?

The trend in the 2021 dataset looks similar to the 2020 dataset. In both datasets, IgG PT levels increase after the booster and reaches its peak around day 14 before slowly decreasing over time. The wP group (teal) generally shows higher responses than the aP group (red) in both datasets. But the 2020 dataset shows more variation (more extreme values) and includes some longer follow up time points (going up to 400 days). The 2021 dataset appears to have a more stable trajectory across patients.

```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost, MFI_normalised, col = infancy_vac, group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept =14, lty = 2)
```



Obtaining CMI-PB RNASeq data

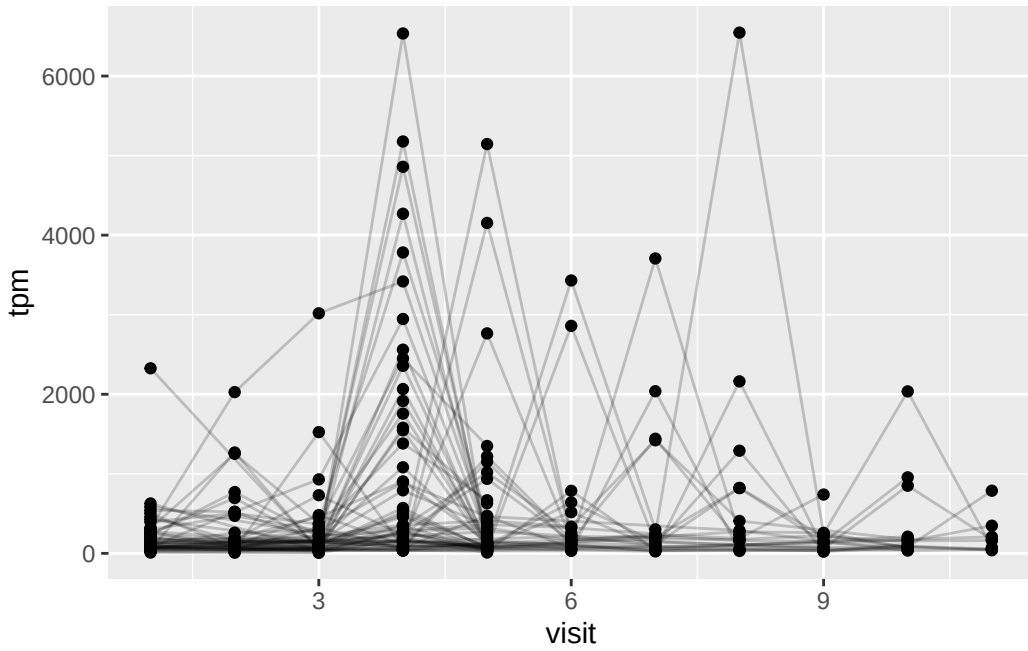
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSG00000211896.7"

rna <- read_json(url, simplifyVector = TRUE)
```

```
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



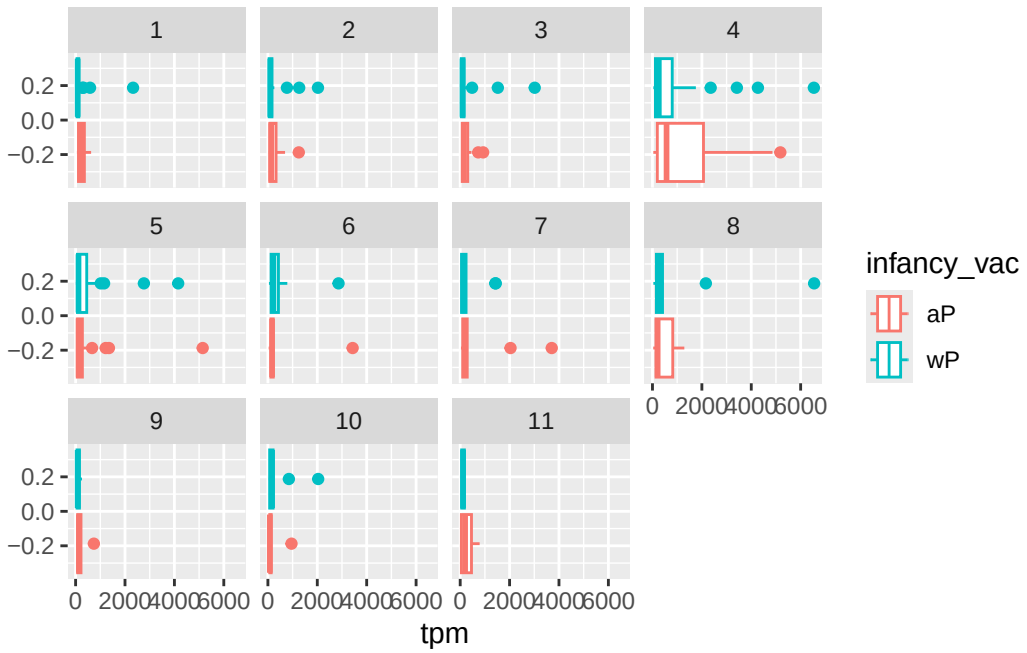
> Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

The gene expression reaches its maximum level around visit 4. The gene expression level is low in the beginning and then it increases dramatically as shown around visit 4 and then it decreases around visits 5-7. It reaches its peak again around visit 8.

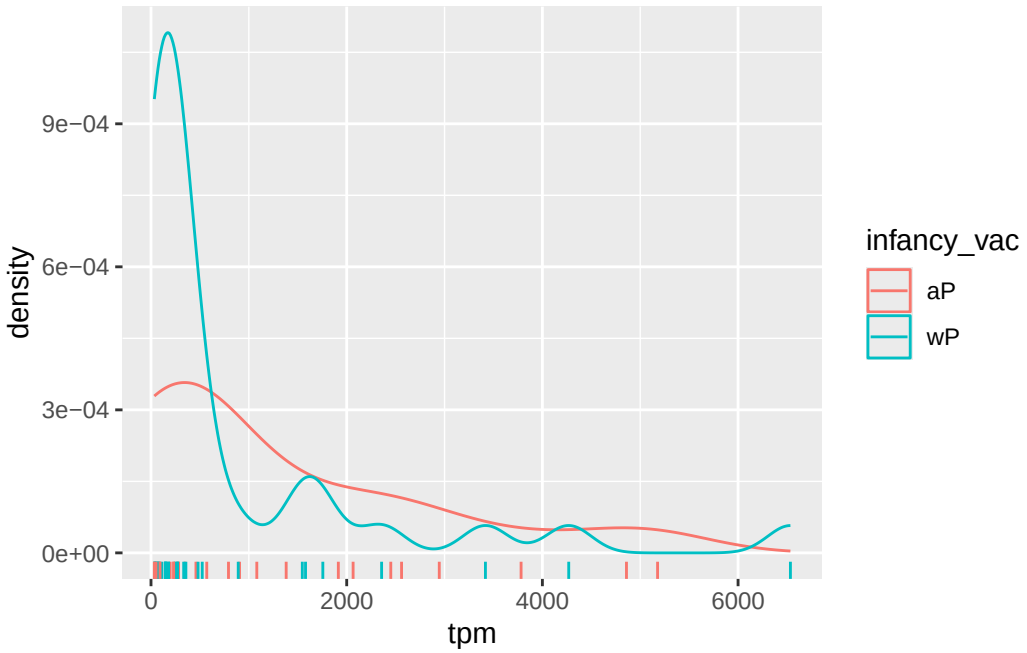
> Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

No, the pattern does not match the trend of antibody titer data. When looking at the earlier antibody plots (PT and FIM2/3), the antibody titers increase after vaccination and peak around the middle visits (visit 5-7) but after the peak, they slowly decrease over time. When looking at the gene expression plot, the gene expression level reaches its peak around visit 4 but then it drops dramatically afterwards. This shows that gene expression changes quickly and peaks earlier while antibody titers increase more slowly and last longer. This is due to gene expression representing early immune activation immediately after vaccination while antibody titers represent the protein products of immune cells which takes time to build up and remains in circulation longer.

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```




```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```




```r
sessionInfo()
```

```
R version 4.4.1 (2024-06-14 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default


locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base
```

```
other attached packages:
[1] dplyr_1.2.0     lubridate_1.9.5 jsonlite_2.0.0  ggplot2_4.0.2

loaded via a namespace (and not attached):
 [1] vctrs_0.7.1        cli_3.6.5          knitr_1.51         rlang_1.1.7
 [5] xfun_0.56          otel_0.2.0         generics_0.1.4     S7_0.2.1
 [9] labeling_0.4.3     glue_1.8.0         htmltools_0.5.9    scales_1.4.0
[13] rmarkdown_2.30     grid_4.4.1         tibble_3.3.1       evaluate_1.0.5
[17] fastmap_1.2.0      yaml_2.3.12        lifecycle_1.0.5    compiler_4.4.1
[21] RColorBrewer_1.1-3 timechange_0.4.0   pkgconfig_2.0.3    rstudioapi_0.18.0
[25] farver_2.1.2       digest_0.6.39      R6_2.6.1           tidyselect_1.2.1
[29] pillar_1.11.1      magrittr_2.0.4     withr_3.0.2        tools_4.4.1
[33] gtable_0.3.6
```